

A Literature-Implied Variant of Remark 13.8 in Argyros–Manoussakis–Pelczar

run fa_banach_001, lane 2

2026-07-18

Status. This is a *literature-implied answer to a strengthened variant*, not a full solution of the source problem. The supporting theorem is Causey’s dualization theorem for the Bourgain c_0 index.

Source problem. Argyros, Manoussakis, and Pelczar ask in Remark 13.8 of [1] whether the following is true. Let X be reflexive and suppose every subspace of X^* has Bourgain c_0 -index greater than ω^ξ . Must there be a subspace $Y \subset X$ such that every subspace of X/Y has Bourgain ℓ_1 -index greater than ω^ξ ?

Supporting theorem. Causey’s Theorem 5.2 in [2] says, in particular, that if a dual space $W = E^*$ has Bourgain c_0 -index greater than ω^ξ , then any predual E has Bourgain ℓ_1 -index greater than ω^ξ ; see also the quantitative paragraph following that theorem.

Literature-implied variant. Let X be reflexive and let $Y \subset X$ be closed. Put $W = Y^\perp \subset X^*$. Assume that every quotient of W has Bourgain c_0 -index greater than ω^ξ . Then every subspace of X/Y has Bourgain ℓ_1 -index greater than ω^ξ .

Proof. Let $Q = X/Y$. Since X is reflexive, Q is reflexive and Q^* is canonically isometric to $Y^\perp = W$. Let $Z \subset Q$ be an infinite-dimensional closed subspace. Its annihilator in Q^* is

$$Z^\perp = \{w \in W : w(z) = 0 \text{ for all } z \in Z\},$$

and the usual duality identification gives

$$Z^* \cong Q^*/Z^\perp \cong W/Z^\perp.$$

By the quotient-hereditary hypothesis on W , this quotient has Bourgain c_0 -index greater than ω^ξ . It is a dual space with predual Z , so Causey’s dualization theorem applies and yields that Z has Bourgain ℓ_1 -index greater than ω^ξ . Since $Z \subset Q$ was arbitrary, $Q = X/Y$ has the desired hereditary ℓ_1 -index conclusion.

In particular, if every quotient of X^* has Bourgain c_0 -index greater than ω^ξ , the same proof with $Y = 0$ gives the desired conclusion for X itself.

Remaining gap for the original question. The original hypothesis is hereditary for subspaces of X^* , whereas the proof above needs quotient-heredity for an annihilator subspace $Y^\perp$. A subspace Z of X/Y produces a quotient $Z^* \cong Y^\perp/Z^\perp$, not merely a subspace of X^* . Thus the unresolved core is the stabilization step: find, from the source hypothesis, an annihilator $Y^\perp \subset X^*$ all of whose quotients still have Bourgain c_0 -index greater than ω^ξ . Causey’s subspace/quotient estimates in Theorem 8.1 are relevant, but they do not alone prove this stabilization.

Bounded search note. The run indexes contained no prior packet for arXiv:0907.4317 or this final problem. Local and web searches for the exact problem phrases and for “dualization of the Bourgain c_0 index” located [2]. No searched source explicitly claimed to resolve Remark 13.8 under the original subspace-hereditary hypothesis.

References

- [1] S. A. Argyros, A. Manoussakis, and A. M. Pelczar, *On the hereditary proximity to ℓ_1* , J. Funct. Anal. 261 (2011), 1145–1203; arXiv:0907.4317.
- [2] R. M. Causey, *Proximity to ℓ_p and c_0 in Banach spaces*, J. Funct. Anal. 269 (2015), 3952–4005; arXiv:1502.05753.